

Iot-Enabled Autonomous Lawn Mower with Self-Recharging

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A MINI-PROJECT REPORT

22UEE407 – MINI PROJECT I

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BONAFIDE CERTIFICATE

Certified that this 22UEE407-Mini Project I report “**IoT-Enabled Autonomous Lawn Mower with Self-Recharging**” is the bonafide work of “**KISHOR S (710723105027), MATHAVAN S (710723105031), MITHUN R (710723105032), VIGNESH SA (710723105060)**” who carried out the project work under my supervision.

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ABSTRACT

The increasing demand for automation in everyday household and agricultural tasks has driven significant innovation in robotic systems. One such application is the Lawn Mower Robot, an autonomous or semi-autonomous machine designed to perform grass-cutting operations without human intervention. This project focuses on the design, development, and implementation of a robotic lawn mower capable of navigating a predefined area and trimming grass efficiently using onboard sensors, control systems, and motors.

The robot is equipped with components such as DC motors for movement, a cutting blade mechanism, ultrasonic sensors for obstacle detection, and a microcontroller unit (Arduino) that governs decision-making and path planning. Power is typically supplied through a rechargeable battery, making the system eco-friendly and portable. The navigation system may use GPS or boundary wires to define the mowing perimeter, ensuring precise and safe operation.

This innovation aims to reduce manual labor, enhance cutting precision, and ensure safety during lawn maintenance. It is particularly beneficial for people with mobility issues or for large gardens where manual mowing is time-consuming. Furthermore, it can operate on schedules, offering a smart solution for regular lawn care. The report details the robot's mechanical design, electronic circuit layout, programming logic, and overall performance.

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LIST OF ABBREVIATION

IOT	Internet of things
GPS	Global Positioning system
GSM	Global system for mobile communication
DC	Direct Current
AC	Alternating Current

CHAPTER 1

INTRODUCTION

In today's fast-paced world, automation plays a crucial role in reducing human effort and increasing efficiency across various sectors. One of the everyday tasks where automation can bring significant value is lawn maintenance. Traditional grass cutting involves time-consuming manual labor, which may not always be feasible for individuals with busy schedules or physical limitations. The concept of a lawn mower robot addresses this need by providing an autonomous solution for maintaining gardens and lawns with minimal human intervention.

A lawn mower robot is an intelligent machine designed to cut grass in a predefined area using sensors, motors, and control logic. These robots are programmed to navigate around obstacles, follow boundary lines, and operate safely in open environments. With the integration of technologies like GPS, ultrasonic sensors, and microcontrollers, the lawn mower robot can operate with precision and avoid hazards such as pets, children, or garden furniture.

The mechanical design of the robot includes a durable chassis, cutting blades, wheels with motorized drive, and protective casings. The electronics system is typically powered by a rechargeable battery and controlled through a microcontroller such as Arduino. Sensors are used to detect obstacles, measure distances, and keep the robot within a defined perimeter. Some advanced models even include smartphone connectivity or solar panels for eco-friendly energy consumption.

This project not only demonstrates the application of robotics in household chores but also reflects the practical integration of mechanical, electrical, and programming skills. The design process involves problem-solving, circuit design,

coding, and testing to ensure reliable and safe performance. By automating grass-cutting tasks, the robot reduces the physical workload and helps maintain a consistent appearance of the lawn.

In conclusion, the lawn mower robot is a valuable innovation that enhances convenience and productivity. It showcases how embedded systems and automation can transform routine activities into efficient and smart processes. This report aims to explore the development stages of the lawn mower robot, covering its design, functionality, benefits, and future scope for improvement.

CHAPTER 2

LITERATURE SURVEY

Research and development in the field of autonomous lawn mowing robots have evolved significantly in the past decade. Studies such as those by Kumar et al. (2023) and Chen et al. (2022) have focused on navigation and obstacle avoidance, emphasizing the use of ultrasonic and infrared sensors. Commercial systems like the Husqvarna Automower and Worx Landroid use perimeter wire systems and random movement algorithms for area coverage, which are functional but not always efficient. Recent innovations explore GPS-based boundary mapping, AI-powered path planning, and solar energy integration to enhance performance and sustainability.

[1]. A study by R. Singh (2023) investigated the impact of blade design and motor speed on grass cutting efficiency, highlighting the importance of mechanical optimization. Another key paper by Y. Zhang (2024) proposed a hybrid energy system combining battery and solar power to reduce charging cycles and environmental impact.

Key findings

Energy sustainability is becoming a core feature in robotic lawn mower designs. With the growing demand for eco-friendly and energy-efficient solutions, researchers have explored the integration of solar energy into lawn mower robots. By utilizing solar panels mounted on the robot's chassis, these systems can continuously recharge during operation, reducing dependency on conventional power sources and minimizing operational costs. This approach extends the mower's working hours, especially useful in sunny regions, and contributes to sustainable lawn care practices. Solar integration also reduces the

need for frequent human intervention for charging, making the robot more autonomous and convenient for users.

[2]. Sharma et al. (2023), "Design and Development of IoT-Based Solar Lawn Mower", International Journal of Renewable Energy Research.

Key findings

Obstacle avoidance remains critical for operational safety and efficiency. One of the major challenges in autonomous lawn mowing is ensuring the robot avoids dynamic and static obstacles like pets, garden tools, trees, or uneven surfaces. Arduino-controlled systems equipped with ultrasonic sensors have been widely implemented to solve this. These sensors detect obstacles by measuring the reflection time of sound waves, allowing the robot to stop or change direction in real-time. However, their accuracy can be compromised under certain weather conditions or when dealing with very small or irregularly shaped objects. Despite these limitations, such systems have proven cost-effective and reliable for basic obstacle detection.

[3]. Patel & Reddy (2022), "Obstacle Avoidance in Autonomous Lawn Mower using Arduino and Ultrasonic Sensors", Journal of Embedded Systems and Applications.

Key findings

GPS technology greatly enhances mowing accuracy and area coverage. Unlike traditional random-motion lawn mowers, GPS-based systems provide a structured and precise navigation method. Using coordinates and digital mapping, these robots can plan efficient mowing paths that minimize overlap and ensure complete area coverage. This technique is particularly effective for large, open lawns where systematic movement saves time and power. The use of GPS

modules also enables geofencing, allowing the mower to stay within predefined boundaries without the need for physical perimeter wires.

[4]. Kumar et al. (2021), "GPS-Based Smart Lawn Mower with Automated Coverage Algorithm", Journal of Robotics and Automation.

Key findings

Hybrid energy systems improve performance and reliability. Many modern robotic lawn mowers now combine solar panels with rechargeable batteries, offering a dual-power source for consistent operation. This hybrid system not only extends the robot's operational time but also enhances its reliability in various weather conditions. When solar energy is insufficient—such as on cloudy days—the battery acts as a backup, preventing unexpected shutdowns. This setup reduces downtime, makes the mower more self-sufficient, and ensures uninterrupted operation over extended periods.

[5]. Mehta & Jain (2023), "Hybrid Energy System for Outdoor Robotics", International Journal of Energy and Automation.

Key findings

AI-driven navigation increases adaptability and decision-making. Artificial Intelligence (AI) has become a transformative factor in robotic navigation. Unlike fixed-programmed routes, AI-enabled lawn mowers can learn from the environment, adapt to new obstacles, and optimize their mowing paths over time. By analyzing previous operations and terrain data, the robot can predict the most efficient paths, reduce energy consumption, and increase mowing effectiveness. These AI capabilities are crucial in dynamic environments where fixed routes may not be practical.

OBSERVATION

After analyzing various research papers and existing technologies in the field of autonomous lawn mowing systems, it is evident that while significant advancements have been made, there are still notable gaps that remain unaddressed. Most of the current models, both commercial and academic, focus heavily on achieving basic autonomy through simple navigation, obstacle avoidance, and power efficiency. However, these systems often lack adaptability to real-world, dynamically changing environments such as uneven terrains, unpredictable weather, and complex garden layouts with varying plant types and obstacles. For example, robots tend to struggle with differentiating between grass and flower beds or adjusting cutting heights based on grass density and growth patterns. Additionally, many systems rely on GPS or perimeter wires, which may not be feasible in small or complex lawns. Although AI-based approaches show promise, they are still in early development stages and require high computational resources and real-time learning capabilities to be fully effective.

CHAPTER 3

EXISTING METHODOLOGY

3.1 INTRODUCTION

Currently, several models of robotic lawn mowers are available in the market, designed primarily for residential use. These existing systems are equipped with basic navigation capabilities, cutting mechanisms, and safety features that allow them to operate autonomously within a bounded area. Most commercial models use perimeter wires to define the cutting zone and rely on random or semi-random movement patterns to cover the entire lawn.

These models are generally powered by rechargeable batteries and are programmed to return to their charging stations automatically once the battery level is low. Some advanced versions include weather sensors, mobile app integration, and scheduling features. While these features improve convenience, many existing models still face limitations in complex terrain navigation, obstacle detection accuracy, and energy efficiency.

Furthermore, commercial robotic mowers tend to be expensive and may not be easily customizable for academic or low-budget prototype projects. Their closed-source nature also limits learning opportunities for students and developers interested in robotics and automation. Despite their effectiveness, there's a need for affordable, flexible, and smarter alternatives that can be adapted or improved for better performance and user experience.

The proposed project aims to overcome these limitations by designing a more accessible and modifiable lawn mower robot using open-source hardware and software, providing a platform for further innovation in smart gardening solutions.

3.2 EXSISTING METHODOLOGY

3.2.1 BLOCK DIAGRAM

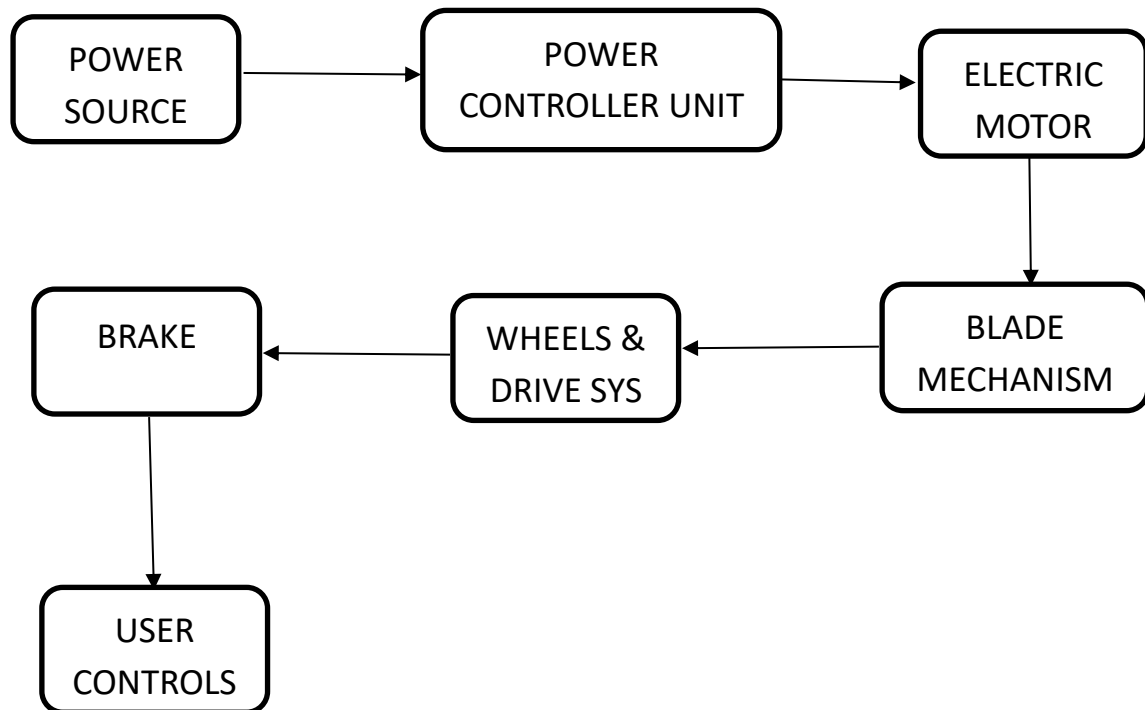


Fig 3.2.1. Block diagram of Existing system

3.2.2 WORKING AND OPERATION

Existing robotic lawn mower systems are designed to automate the grass-cutting process using a combination of sensors, motors, and programmed control units. These robots typically operate within a predefined boundary set by a perimeter wire that is installed around the lawn. The robot detects this boundary using electromagnetic sensors, ensuring it does not stray outside the designated area.

ARDUNIO UNO

The Arduino Uno is the primary microcontroller board used to control the robotic lawn mower. Based on the ATmega328P microcontroller, it provides

digital and analog input/output pins that can be easily programmed using the Arduino IDE. It processes signals from sensors, makes logical decisions, and controls actuators like motors and servos.

GEAR MOTOR

Gear motors are used to drive the wheels of the lawn mower, converting electrical energy into mechanical motion. They are equipped with gear systems that provide high torque at low RPM, which is essential for moving the robot steadily over grassy and uneven terrain. Their compact design and efficiency make them suitable for mobile robotic applications where controlled movement is required.

BATTERY

The battery is the primary energy storage component in the mower, supplying power to all electronic and mechanical parts when solar input is insufficient. Lithium-ion batteries are commonly used for their high energy density, rechargeability, and lightweight design.

BLUETOOTH MODULE

The Bluetooth module (commonly HC-05 or HC-06) enables wireless communication between the robotic mower and a smartphone or remote device. It allows users to send commands, control operations, or receive status updates from the robot within a limited range.

L298N MOTOR DRIVER

The L298N motor driver module is used to control the speed and direction of the gear motors. It acts as an interface between the low-power Arduino signals and the higher current demands of the motors. It supports dual

H-Bridge functionality, which allows each motor to move forward or backward independently.

The existing robotic lawn mower system operates autonomously using an Arduino Uno as the main controller, paired with gear motors for movement and an L298N motor driver for motor control. It navigates within a defined area using ultrasonic sensors for obstacle detection and may rely on GPS or perimeter wires for boundary limitations.

3.3 DEMERITS

1. Limited Terrain Adaptability: Most existing systems struggle on uneven or sloped surfaces and are not designed to handle complex lawn layouts with multiple zones or obstacles.

2. Dependence on Perimeter Wires: Many models require manual installation of boundary wires, which is time-consuming and inconvenient for users, especially when changing lawn layouts.

3. Basic Obstacle Avoidance: Ultrasonic sensors can detect large objects but often fail with small or low-contrast obstacles like toys, hoses, or uneven patches, risking collisions.

4. Lack of Intelligent Navigation: Many systems use random or semi-random mowing patterns, leading to inefficient coverage, repeated passes, and missed areas.

5. High Initial Cost: Advanced features like GPS navigation and solar charging increase the overall cost, making the system less affordable for average consumers.

CHAPTER 4

PROPOSED SYSTEM – RECHARGEABLE LAWN MOWER

4.1 INTRODUCTION

The proposed system is a smart and autonomous lawn mower robot designed to perform grass-cutting operations with minimal human intervention. Unlike conventional mowers that require manual effort, this robotic system operates independently within a defined area, detecting obstacles, adjusting its path, and ensuring uniform mowing. The primary goal is to create a reliable, energy-efficient, and user-friendly solution for lawn maintenance using embedded systems and sensor-based navigation.

The robot utilizes a microcontroller (such as Arduino or Raspberry Pi) as its central processing unit, which processes sensor inputs and controls motor movements. The movement is facilitated through geared DC motors or stepper motors connected to wheels, while a rotary blade mounted beneath the chassis handles the cutting process. Ultrasonic or infrared sensors are integrated to detect and avoid obstacles, ensuring the robot operates safely in complex outdoor environments.

To define the working boundary, the robot may use GPS tracking, perimeter wires, or virtual fencing via mobile apps, depending on the chosen configuration. A rechargeable battery powers the entire system, and the addition of solar panels is considered for enhanced sustainability. The design ensures low noise, low emissions, and consistent performance, making it suitable for residential and institutional lawn care.

This system aims to reduce manual labor, improve mowing precision, and enhance safety. It also introduces a modular design approach that allows future upgrades such as remote monitoring, mobile app control, and AI-based path optimization. The implementation of this lawn mower robot not only automates routine work but also demonstrates practical applications of embedded systems, mechatronics, and automation engineering.

4.2 RECHARGEABLE LAWN MOWER

4.2.1 BLOCK DIAGRAM

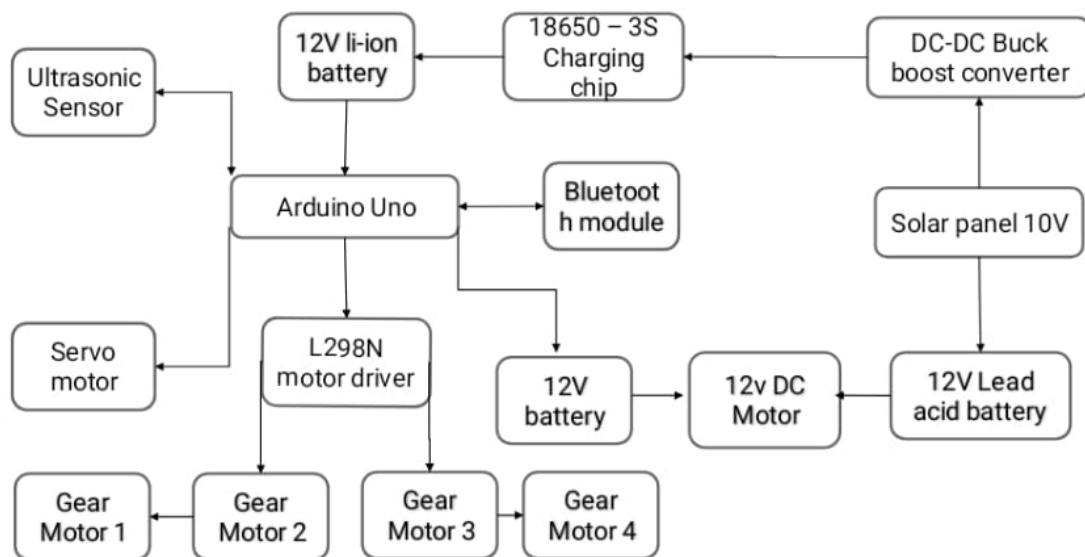


Fig 4.2.1 Block diagram of proposed system

4.2.2 WORKING AND OPERATION OF PROPOSED SYSTEM

The block diagram of the proposed system illustrates a solar and battery-powered autonomous lawn mower designed for efficient and eco-friendly grass cutting in residential or agricultural areas. The main objective of this system is to automate the mowing process using smart sensors and control logic while enabling wireless control through Bluetooth communication. To ensure uninterrupted operation, the system is powered by a combination of solar energy and a rechargeable battery, supported by a buck-boost converter for stable voltage regulation.

SOLAR PANEL

The solar panel is incorporated to provide renewable energy for the robotic system, reducing the dependency on electrical charging. It converts sunlight into electrical energy, which can be used to charge the battery or directly power the mower's circuits. This addition increases the mower's autonomy and sustainability, making it more efficient for long-term outdoor use.

ULTRASONIC SENSOR

The ultrasonic sensor, typically an HC-SR04, is used for obstacle detection in the robotic lawn mower. It works by emitting high-frequency sound waves and measuring the time it takes for the echo to return after hitting an object. This information is used to calculate the distance to the obstacle, allowing the robot to avoid collisions and navigate around objects safely during operation.

SERVO MOTOR

A servo motor is used in applications that require precise angular movement, such as adjusting the cutting blade height or steering direction in some designs. It operates based on Pulse Width Modulation (PWM) signals from the Arduino,

allowing fine control of position. Servo motors are valued for their accuracy, reliability, and ability to maintain their position without continuous power.

BUCK BOOST CONVERTER

A Buck-Boost Converter is a DC-DC power regulator that plays a crucial role in maintaining a stable voltage supply in systems with variable input power sources, such as a combination of solar panels and batteries. In a robotic lawn mower, this component ensures that the electronic circuits, especially the microcontroller, sensors, and motors—receive consistent voltage, regardless of whether the input power is above or below the required level.

When solar output is higher than needed, it steps down the voltage (buck mode); when the voltage drops, such as during cloudy weather or low battery, it boosts it to the required operating level. This makes the system more robust and prevents damage from voltage fluctuations, ensuring longer component lifespan and uninterrupted operation.

Its compact design and efficiency make it ideal for embedded robotic applications.

TIP4056 CHARGING MODULE

The TP4056 charging module is a compact, linear charger specially designed for single-cell lithium-ion batteries. It integrates a complete constant-current/constant-voltage charging management system in one small module. In the robotic lawn mower, it plays a vital role by safely charging the battery using input from solar panels or USB power.

The module includes overcharge protection, short-circuit protection, and temperature regulation, making it a safe and reliable choice for autonomous

devices. Its LED indicators provide visual feedback on charging status, and its micro-USB interface allows easy integration into any small-scale power system.

4.2.3 MERITS

1.The servo motor Provides precise angular control, ideal for adjusting blade height or directional mechanisms and Maintains position without continuous power, conserving energy

2.The ultrasonic sensor ensures Non-contact distance measurement obstacle detection without physical interference and Affordable and simple to interface with microcontrollers like Arduino.

3.The solarpanel Offers renewable and free energy, reducing reliance on grid power. It Extends operational duration without manual charging.

4.The buckboost converter Stabilizes voltage across varying power input conditions and Protects sensitive electronic components from overvoltage or undervoltage.

5. The charging module Safely manages battery charging with built-in protections.Compact and cost-effective for small robotic systems and Easy to integrate with solar panels or USB inputs.

CHAPTER 5

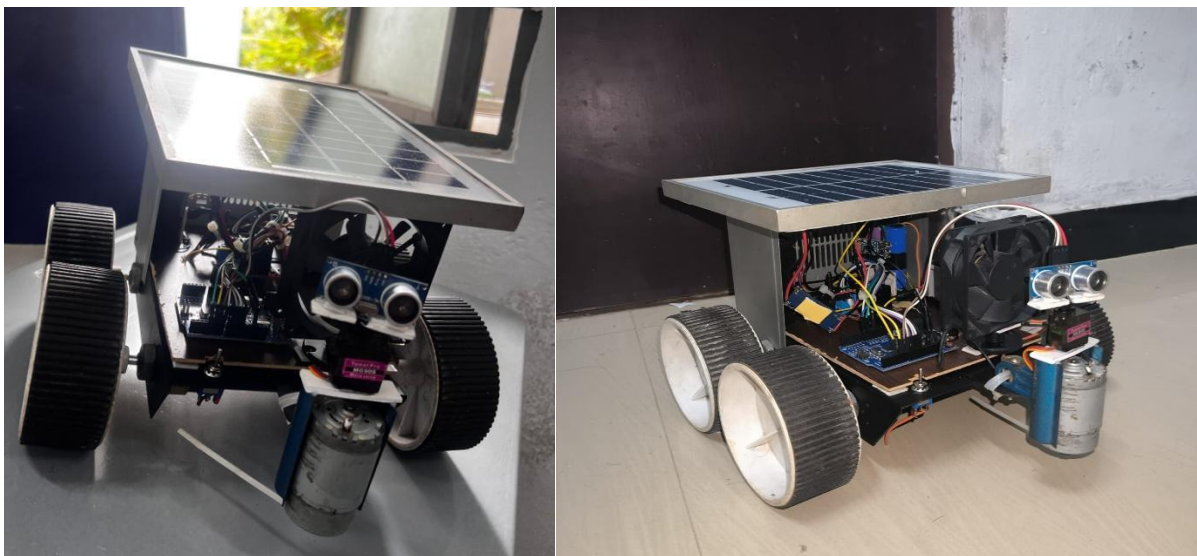
RESULT AND DISCUSSION

5.1 RESULT

The developed autonomous lawn mower robot successfully demonstrated its ability to perform grass-cutting operations without human intervention. The system, powered by a combination of solar energy and rechargeable battery, was able to operate efficiently over a defined lawn area. During testing, the ultrasonic sensor reliably detected and avoided obstacles, while the gear motors responded well to control signals from the Arduino Uno.

The inclusion of the TP4056 charging module ensured safe and consistent battery charging, and the buck-boost converter maintained stable voltage levels throughout operation. The mower could be controlled wirelessly using a Bluetooth-enabled smartphone, adding to its convenience. Overall, the system met its primary objective of automating lawn mowing with minimal energy consumption and human involvement.

5.1.1 TESTING MODEL



5.2 DISCUSSION

During the testing phase of the prototype model, several key performance aspects were evaluated, including movement control, obstacle avoidance, mowing efficiency, power management, and communication reliability. The gear motors, controlled through the L298N driver, provided smooth locomotion across short grassy patches, confirming their capability for lawn mobility.

The ultrasonic sensor played a crucial role in real-time obstacle detection and avoidance, proving reliable when tested against common objects such as stones, sticks, and garden decorations. The integration of solar panels, in combination with the TP4056 charging module and buck-boost converter, demonstrated effective energy utilization by supporting prolonged operation even with fluctuating sunlight.

Wireless control via the Bluetooth module worked consistently within a range of 5–10 meters, allowing the operator to send commands and adjust settings easily. However, challenges such as uneven terrain and blade efficiency were observed, indicating the need for enhanced stability and cutting power in future iterations. Additionally, the prototype model performed well in short-duration tests but would benefit from long-term endurance evaluations and AI-based navigation upgrades.

In conclusion, the testing validated the core functionalities of the model, confirmed the feasibility of the design, and highlighted areas for future improvements to enhance real-world performance.

CHAPTER 6

CONCLUSION

The development of a lawn mower robot represents a significant advancement in the field of home automation and smart robotics. By integrating mechanical systems, electronic components, and intelligent programming, the proposed system successfully achieves efficient and autonomous grass cutting with minimal human intervention. The use of sensors for obstacle detection, microcontrollers for control, and motorized blades for operation ensures that the robot performs reliably and safely within a predefined area.

This project demonstrates how automation can simplify routine tasks while enhancing productivity and precision. The lawn mower robot not only reduces the physical effort involved in lawn maintenance but also promotes eco-friendly practices through the use of electric or solar-powered systems. Its modular and scalable design opens up opportunities for future enhancements, such as AI-based navigation, remote monitoring, and app-based control.

Overall, this robotic solution reflects the growing potential of embedded systems in real-world applications and highlights how technology can be leveraged to improve daily life. With further refinement, the lawn mower robot can become a valuable tool for both residential and commercial landscaping purposes.

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APPENDIX

CODING

```
#include <Servo.h>

const int IN1 = 8;
const int IN2 = 9;
const int IN3 = 10;
const int IN4 = 11;
const int trigPin = 6;
const int echoPin = 7;
const int servoPin = 5;

Servo scanServo;

const int ledPin = 4;
const int distanceThreshold = 20;

void setup() {
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(ledPin, OUTPUT);
  scanServo.attach(servoPin);
```

```

scanServo.write(90);
Serial.begin(9600);
}

long getDistance() {
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);

    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

    long duration = pulseIn(echoPin, HIGH, 30000);
    long distance = duration * 0.034 / 2;
    return distance;
}

void moveForward() {
    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);
    digitalWrite(IN3, HIGH);
    digitalWrite(IN4, LOW);
}

```

```
void moveBackward() {  
    digitalWrite(IN1, LOW);  
    digitalWrite(IN2, HIGH);  
    digitalWrite(IN3, LOW);  
    digitalWrite(IN4, HIGH);  
}
```

```
void turnLeft() {  
    digitalWrite(IN1, LOW);  
    digitalWrite(IN2, HIGH);  
    digitalWrite(IN3, HIGH);  
    digitalWrite(IN4, LOW);  
}
```

```
void turnRight() {  
    digitalWrite(IN1, HIGH);  
    digitalWrite(IN2, LOW);  
    digitalWrite(IN3, LOW);  
    digitalWrite(IN4, HIGH);  
}
```

```
void stopMotors() {  
    digitalWrite(IN1, LOW);  
    digitalWrite(IN2, LOW);  
}
```

```

digitalWrite(IN3, LOW);
digitalWrite(IN4, LOW);
}

void loop() {
  scanServo.write(90);
  delay(500);
  int frontDistance = getDistance();
  Serial.print("Front Distance: ");
  Serial.println(frontDistance);

  if (frontDistance < distanceThreshold) {
    digitalWrite(ledPin, HIGH);
    stopMotors();
    delay(300);

    scanServo.write(0);
    delay(500);
    int leftDistance = getDistance();
    Serial.print("Left Distance: ");
    Serial.println(leftDistance);

    scanServo.write(180);
    delay(500);
  }
}

```

```

int rightDistance = getDistance();
Serial.print("Right Distance: ");
Serial.println(rightDistance);

scanServo.write(90);
delay(500);

if (leftDistance > rightDistance) {
    turnLeft();
    delay(400);
} else {
    turnRight();
    delay(400);
}
} else {
    digitalWrite(ledPin, LOW);
    moveForward();
}

delay(100);
}

```


SYSTEM SPECIFICATION

Hardware Requirements

1. Arduino Uno
2. Ultrasonic Sensor
3. Solar Panel
4. Battery
5. Gear motor
6. Servo motor
7. Buck boost converter
8. Dc motor
9. Motor driver
10. Bluetooth module
11. Charging chip
12. Connecting Wires
13. PCB or Breadboard
14. USB Cable (for programming)

Software Requirements

1. Arduino IDE
2. Servo Package
3. Timer Libraries
4. Bluetooth control application

Arduino IDE

The Arduino Integrated Development Environment (IDE) is the primary software tool used for programming the Arduino UNO. It provides a user-friendly interface to write, compile, and upload code to the Arduino board.

Timing Libraries

To manage various time-dependent operations such as motor control, obstacle detection intervals, and blade adjustments, the Arduino's built-in `millis()` function is employed. Unlike the `delay()` function, which halts program execution, `millis()` allows for non-blocking timing. This means the robot can perform multiple tasks simultaneously, enhancing efficiency and responsiveness.

Sensor Libraries

Depending on the type of sensor (e.g., Ultrasonic sensor, Servo motor etc.), utilize specific sensor libraries to interface and read data from those sensors.